

## Evaluating Limits

Date\_\_\_\_\_ Period\_\_\_\_

**Evaluate each limit.**

1)  $\lim_{x \rightarrow 0} -1$

2)  $\lim_{x \rightarrow -4} (-x - 4)$

3)  $\lim_{x \rightarrow -2} (2x + 3)$

4)  $\lim_{x \rightarrow 3} (-x^3 + 13x^2 - 56x + 81)$

5)  $\lim_{x \rightarrow 2} (x^3 - 2x^2 - 2)$

6)  $\lim_{x \rightarrow 5} -\sqrt{2x + 1}$

7)  $\lim_{x \rightarrow -2} \sqrt[3]{-2x - 3}$

8)  $\lim_{x \rightarrow -3} \frac{x - 3}{x^2 + 2x + 2}$

9)  $\lim_{x \rightarrow 2} \frac{x - 6}{x^2 - 4x + 3}$

10)  $\lim_{x \rightarrow \frac{\pi}{6}} \cos(x)$

**Critical thinking questions:**

11) Give an example of a limit that evaluates to 5.

12) Give an example of a limit of a quadratic function where the limit evaluates to 16.